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Class: - B.Sc. (ECS)-II Div: - A, B
Home Assignment: - FCFS (First-Come, First-Served).

- **FCFS:** The process that arrives first gets executed first.
- **Non-preemptive:** Once a process starts executing, it runs till completion.
 - **Arrival Time (AT):** When the process arrives in the queue.
 - **Burst Time (BT):** The time it takes to execute the process.
 - **Completion Time (CT):** When the process finishes execution **CT=AT+WT+BT.**
 - **Turnaround Time (TAT) = CT - AT**
 - **Waiting Time (WT) = TAT – BT**
 - **Response time = first response time- arrival time.**

Calculate the following terms Completion Time, Turnaround Time, Waiting Time, Response time, Average waiting and Average turnaround time.

1. Example 1 – Basic with 3 processes

Process	AT	BT
P1	0	4
P2	1	3
P3	2	1

2. Example 2 – All arrive at same time

Process	AT	BT
P1	0	2
P2	0	5
P3	0	3

3. Example 3 – One long process first

Process	AT	BT
P1	0	10
P2	2	2
P3	3	1

4. Example 4 – Idle time at start

Process	AT	BT
P1	3	4
P2	5	2
P3	6	1

5. Example 5 – Processes arrive at different intervals

Process	AT	BT
P1	0	2
P2	2	4
P3	5	3

6. Example 6 – Processes with increasing burst time

Process	AT	BT
P1	0	1
P2	1	2
P3	2	3

7. Example 7 – Processes with decreasing burst time

Process	AT	BT
P1	0	3
P2	1	2
P3	2	1

8. Example 8 – Identical burst times

Process	AT	BT
P1	0	4
P2	1	4
P3	2	4

9. Example 9 – One process arrives much later

Process	AT	BT
P1	0	3
P2	1	2
P3	10	1

10. Example 10 – Two processes arrive at same time

Process	AT	BT
P1	0	5
P2	0	2
P3	1	3

11. Example 11 – All arrive late

Process	AT	BT
P1	5	2
P2	6	4
P3	7	1

12. Example 12 – More than 3 processes

Process	AT	BT
P1	0	2
P2	1	1
P3	2	3
P4	3	2

13. Example 13 – Long queue, increasing AT

Process	AT	BT
P1	0	3
P2	2	2
P3	4	4
P4	6	1
P5	8	2

14. Example 14 – Random arrival times and burst

Process	AT	BT
P1	1	3
P2	0	2
P3	4	1
P4	3	2

15. Example 15 – All processes arrive after time 5

Process	AT	BT
P1	5	1
P2	6	3
P3	7	2